

## Python for Data Science - 2305CS303

### Lab - 3

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#### 1. WAP to print 1 to 10.

```
In [1]: for i in range(1,11):  
        print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

#### 2. WAP to print 1 to n.

```
In [3]: n=int(input("enter n: "))  
        for i in range(1,n+1):  
            print(i)
```

```
enter n: 10  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

### 3.WAP to print odd numbers between 1 to n.

```
In [2]: n=int(input("enter n: "))
        for i in range(1,n+1,2):
            print(i)
```

enter n: 10

1  
3  
5  
7  
9

### 4. WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

```
In [4]: n1=int(input("enter n: "))
        n2=int(input("enter n2: "))
        for i in range(n1,n2+1):
            if i%2==0 and i%3!=0:
                print(i)
```

enter n: 10

enter n2: 20

10  
14  
16  
20

### 5. WAP to print sum of 1 to n numbers.

```
In [8]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
            sum+=i
        print(sum)
```

enter n: 10

55

### 6.WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

```
In [10]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
            sum+=i*i
        print(sum)
```

enter n: 2

5

### 7. WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

```
In [11]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
```

```

    if i%2==0:
        sum-=i
    else:
        sum+=i
print(sum)

```

enter n: 7  
4

## 8. WAP to print multiplication table of given number.

```

In [12]: n1=int(input("enter n: "))
        for i in range(1,11):
            print(f"{n1} * {i}= {n1*i}")

```

enter n: 10  
10 \* 1= 10  
10 \* 2= 20  
10 \* 3= 30  
10 \* 4= 40  
10 \* 5= 50  
10 \* 6= 60  
10 \* 7= 70  
10 \* 8= 80  
10 \* 9= 90  
10 \* 10= 100

## 9. WAP to find factorial of the given number.

```

In [13]: n1=int(input("enter n: "))
        sum=1
        for i in range(n1,1,-1):
            sum*=i
        print(sum)

```

enter n: 5  
120

## 10. WAP to find factors of the given number.

```

In [14]: n1=int(input("enter n: "))
        for i in range(1,n1+1):
            if n1%i==0:
                print(i)

```

enter n: 10  
1  
2  
5  
10

## 11. WAP to find whether the given number is prime or not.

```

In [16]: n1=int(input("enter n: "))
        # flag=True
        for i in range(2,n1):
            if n1%i==0:
                print("not prime number")

```

```
        break
    else:
        print("prime number")
```

enter n: 7  
prime number

## 12. WAP to print sum of digits of given number.

```
In [18]: n1=int(input("enter n: "))
sum1=0
while n1>0:
    digit=n1%10
    sum1+=digit
    n1//=10
print(int(sum1))
```

enter n: 123  
6

## 13. WAP to check whether the given number is palindrome or not.

```
In [37]: n1=input("enter n: ")
n2=int(n1[::-1])
n1=int(n1)
if n1==n2:
    print("palindrome")
```

enter n: 121  
palindrome

## 14. Fibonacci Number

```
In [64]: n1=int(input("enter n: "))
a, b = 0, 1
for i in range(n1):
    print(a)
    a, b = b, a + b
```

enter n: 5  
0  
1  
1  
2  
3

## 15. Armstrong Number

```
In [73]: n1=int(input("enter n: "))
l=len(str(n1))
sum=0
t=n1
while n1>0:
    div=n1%10
    sum+= div**l
    n1//=10
if sum == t:
```

```
    print("it is Armstrong num")
else:
    print("not armstrong num")
```

enter n: 153  
it is Armstrong num

## 16. Perfect number

```
In [77]: n1=int(input("enter n: "))
sum=0
for i in range(1,n1):
    if n1%i==0:
        sum+=i
if sum==n1:
    print("perfect number")
else:
    print("not perfect number")
```

enter n: 496  
perfect number

## 17. GCD

```
In [80]: n1=int(input("enter n: "))
n2=int(input("enter n2: "))
for i in range(1,min(n1,n2)):
    if n1%i==0 and n2%i==0:
        max=i
print(max)
```

enter n: 8  
enter n2: 16  
4

## 18. LCM

```
In [81]: n1=int(input("enter n: "))
n2=int(input("enter n2: "))
for i in range(1,min(n1,n2)):
    if n1%i==0 and n2%i==0:
        max=i
print((n1*n2)//max)
```

enter n: 6  
enter n2: 8  
24

## 19. count no of digit

```
In [83]: n1=int(input("enter n: "))
print(len(str(n1)))
```

enter n: 456654456654  
12

## 20.largest digit

```
In [86]: num = int(input("enter n: "))  
         digits = list(map(int, str(num)))  
         digits.sort()  
         print(digits[-1])
```

enter n: 123

3